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Adaptive immunity in the etiology and progression of Parkinson's disease

Gregory Williams, Antoine Freuchet, April Frazier, Cecilia Lindestam Arlehamn

[La Jolla Institute for Immunology, La Jolla, CA, USA]

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Summary (or Abstract)

Immune cell population frequencies and antigen-specific T cell reactivity in PBMC were studied using spectral flow cytometry and Fluorospot assays.

For flow cytometry data, a multi-color panel was developed and used to capture several populations and subpopulations including monocytes, NK cells, and T cells. For Fluorospot data, cytokine values of interferon-gamma, IL-5, and IL-10 were analyzed to determine the magnitude of T cell responses towards alpha-synuclein, Epstein-Barr virus (control), and Pertussis (control) derived from the peripheral blood of donors.

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Method

Peripheral blood mononuclear cells (PBMCs) were thawed, washed, counted, and then split for use in the two different assays (flow cytometry, Fluorospot).

For flow cytometry, cells were washed in PBS, dead cells were stained using Fixable Viability Dye eFluor506 (Thermo Fisher) for 20 min at 4°C. Cells were washed and blocked with Fc block (BD Biosciences) for 15 min at 4°C to block unspecific antibody binding. Cells were washed again and a cocktail of staining antibodies was added for 20 min at 4°C. Cells were acquired on a Spectral Aurora (Cytek), .fcs files were analyzed in FlowJo software (version 10.8.1), cells were gated on live, CD45+ singlets, and then subsequently phenotyped based off the fluorophore-conjugated antibodies to measure various immune cell subset frequencies in the peripheral blood.

For Fluorospot, cells were co-cultured with alpha-synuclein, Epstein-Barr virus, and Pertussis peptide pools and subsequently expanded *in vitro* over a two-week period. Cultures were supplemented with IL-2 (10 U/mL) every 4 days. After two weeks, expanded cells were harvested and were re-stimulated in triplicates with the same peptide pools, dimethylsulfoxide (DMSO; as a negative control) and phytohemagglutinin (PHA; as a positive control) and then processed for a tri-color (interferon-gamma, IL-5, and IL-10) Fluorospot assay (Mabtech). Spot forming cells (SFC) were captured using an IRIS Fluorospot reader (Mabtech) and results were analyzed using Excel (Microsoft). To be considered positive, antigen-specific T cells had >100 SFC per million PBMC, p-value <0.05 and >2 fold increase when compared to the DMSO control. Note: "ND" refers to "no data", as some samples either did not have the adequate amount of cells for testing or did not pass our SFC quality control criteria.

References

About the Authors

This document was prepared by **Gregory Williams, Antoine Freuchet, April Frazier, and Cecilia Lindestam Arlehamn, La Jolla Institute for Immunology, Center for Infectious Disease and Vaccine Research**. For more information, please contact **Cecilia Lindestam Arlehamn** at **(858) 876-4754** or by email at cecilia@lji.org.